

Unit - V Trends in Cloud

Weightage marks

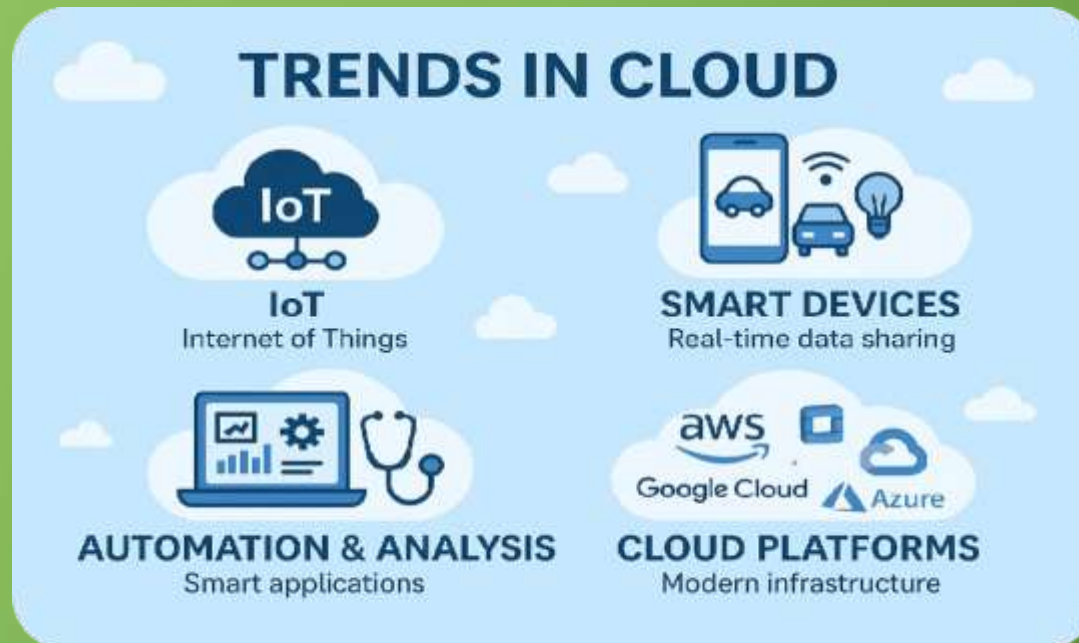
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Syllabus

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.
5	TLO 5.1 Explain the characteristics of the enabling technology with the IoT. TLO 5.2 Select relevant Cloud platform or application for development. TLO 5.3 Describe the features of Cloud-based smart device. TLO 5.4 Compare features of various Cloud platforms.	Unit - V Trends in Cloud 5.1 Cloud trends in supporting Ubiquitous Computing 5.2 Enabling Technology in the Internet of Things(RFID, Sensor Networks and ZigBee Technologies, GPS) 5.3 Innovative Applications with the Internet of Things (Ex: Health care: ECG Analysis in Cloud and it's access, CRM and ERP: Business and Consumer Application) 5.4 Benefits of Cloud Platforms : Amazon EC2 and S3, CloudStack, Intercloud, Google App Engine, Open stack, Open Nebulla

Trends in Cloud -Introduction

- Cloud computing is growing fast with new trends like IoT, ubiquitous computing, and smart devices. These trends help in real-time data sharing, automation, and smart applications. Cloud platforms also support modern needs in healthcare, business, and more using technologies like RFID, sensors, and GPS.



5.1 Cloud trends in supporting Ubiquitous Computing

Possible Questions

- What is ubiquitous computing? How does cloud support it? (4M)
- Explain the trends of cloud in smart city or smart home use. (4M)
- Write any four features of ubiquitous computing. (2M)

What is Ubiquitous Computing?

- Also called **pervasive computing**.
- Means “**computing everywhere and anytime**”.
- Devices and applications work in the background and are always connected.



Applications / Examples

- **Smart Homes** – Lights, fans, and appliances controlled remotely.
- **Smart Cities** – Traffic control, public Wi-Fi, waste management.
- **Healthcare** – Remote patient monitoring, ECG in cloud.
- **Education** – Cloud-based e-learning platforms

Role of Cloud in Ubiquitous Computing

1. Anytime Access – Users can access data from anywhere at any time.
2. Remote Control – Devices can be controlled from mobile or web interfaces.
3. Data Storage – Stores huge amounts of data from smart devices.
4. Real-time Communication – Sends and receives data instantly.
5. Scalability – Easily supports growing number of users and devices.

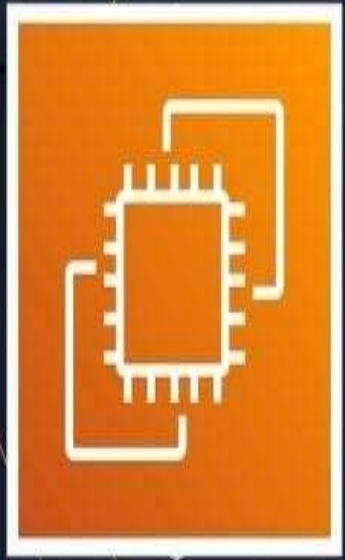
Enlist any 4 cloud platform . (2M)

1. Amazon Web Services (AWS)
2. Microsoft Azure
3. Google Cloud Platform (GCP)
4. IBM Cloud
5. Oracle Cloud
6. Alibaba Cloud
7. Salesforce Cloud
8. DigitalOcean
9. Red Hat OpenShift
10. VMware Cloud

5.4 Benefits of Cloud Platforms

- Cloud platforms provide a base to develop, run, and manage cloud-based applications and services. These platforms offer **storage, scalability, flexibility, and cost-efficiency**.

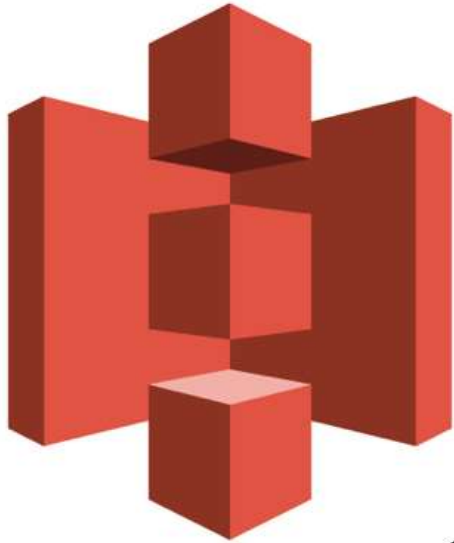
1. Amazon EC2 (Elastic Compute Cloud)



**Amazon
EC2**

- Provides virtual servers (instances).
- Scalable computing capacity.
- Pay-as-you-use pricing.
- Easy to deploy and manage apps.

2. Amazon S3 (Simple Storage Service)



Amazon S3

- Provides secure object storage.
- Highly scalable and durable.
- Used for backup, websites, big data, etc.
- Supports versioning and encryption.

3. CloudStack



- Open-source cloud management platform.
- Manages large-scale cloud infrastructure.
- Supports multi-tenancy.
- User-friendly interface and automation.

4. Intercloud



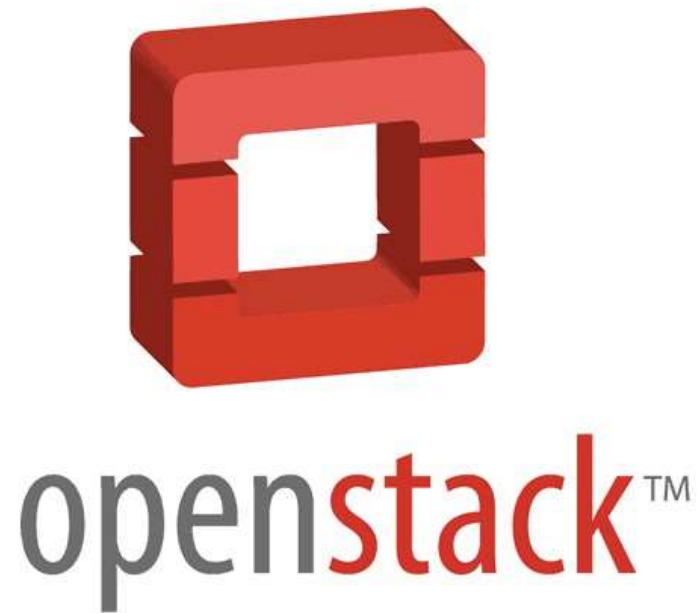
- Connects multiple cloud services.
- Improves data availability and redundancy.
- Reduces dependency on one provider.
- Supports interoperability

5. Google App Engine



- Platform as a Service (PaaS).
- Supports multiple programming languages.
- Automatically scales apps.
- Developers don't manage infrastructure.

6. OpenStack



- Open-source cloud software.
- Manages compute, storage, and networking.
- Scalable and flexible.
- Used in private and public clouds.

7. OpenNebula



- Lightweight cloud platform.
- Easy for private and hybrid cloud setup
- Open-source and customizable.
- Simple interface and fast deployment.

Questions (HOMEWORK) Msbte asked!!

- Describe the working of Google App Engine. (4m) / Describe Google App Engine. Explain what kind of problem can be solved using Google App Engine (4m)
- Describe Amazon EC2 . (4m)
- Differentiate between OpenNebula and Openstack. (4M)
- Explain working of Amazon EC2 & S3 cloud platform (4M)
- Differentiate between open stack & Eucalyptus. (4M)
- Explain how Identity management can be implement using open stack. (4m)

5.2 Enabling Technology in the Internet of Things(RFID, Sensor Networks and ZigBee Technologies, GPS)

- Enabling technologies are the core technologies that support the implementation of IoT systems. These technologies help in identifying, sensing, communicating, and locating things (devices or objects).
- Enabling technologies are the core components that make IoT systems work. These include:
 1. RFID (Radio Frequency Identification)
 2. Sensor Networks
 3. ZigBee Technology
 4. GPS (Global Positioning System)

1. RFID (Radio Frequency Identification):

- RFID is a **wireless technology** that uses **radio waves** to identify and track objects automatically using **tags** and **readers**.

Components:

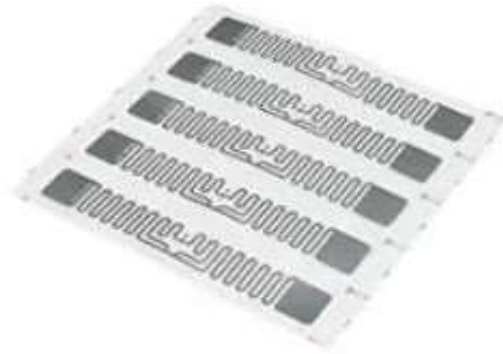
- **RFID Tag** - Stores data about the object.
- **RFID Reader** - Reads data from the tag.
- **Antenna** - Transmits signals between tag and reader.



Types of RFID Tags



Active RFID



Passive RFID



BAP RFID

Types of Tags:

- Passive Tag - No internal power source, low range.
- Active Tag - Has battery, long range.

Features and Application

- Features:

- Contactless communication.
- Works without line-of-sight.
- Can read multiple tags at once.

- Applications:

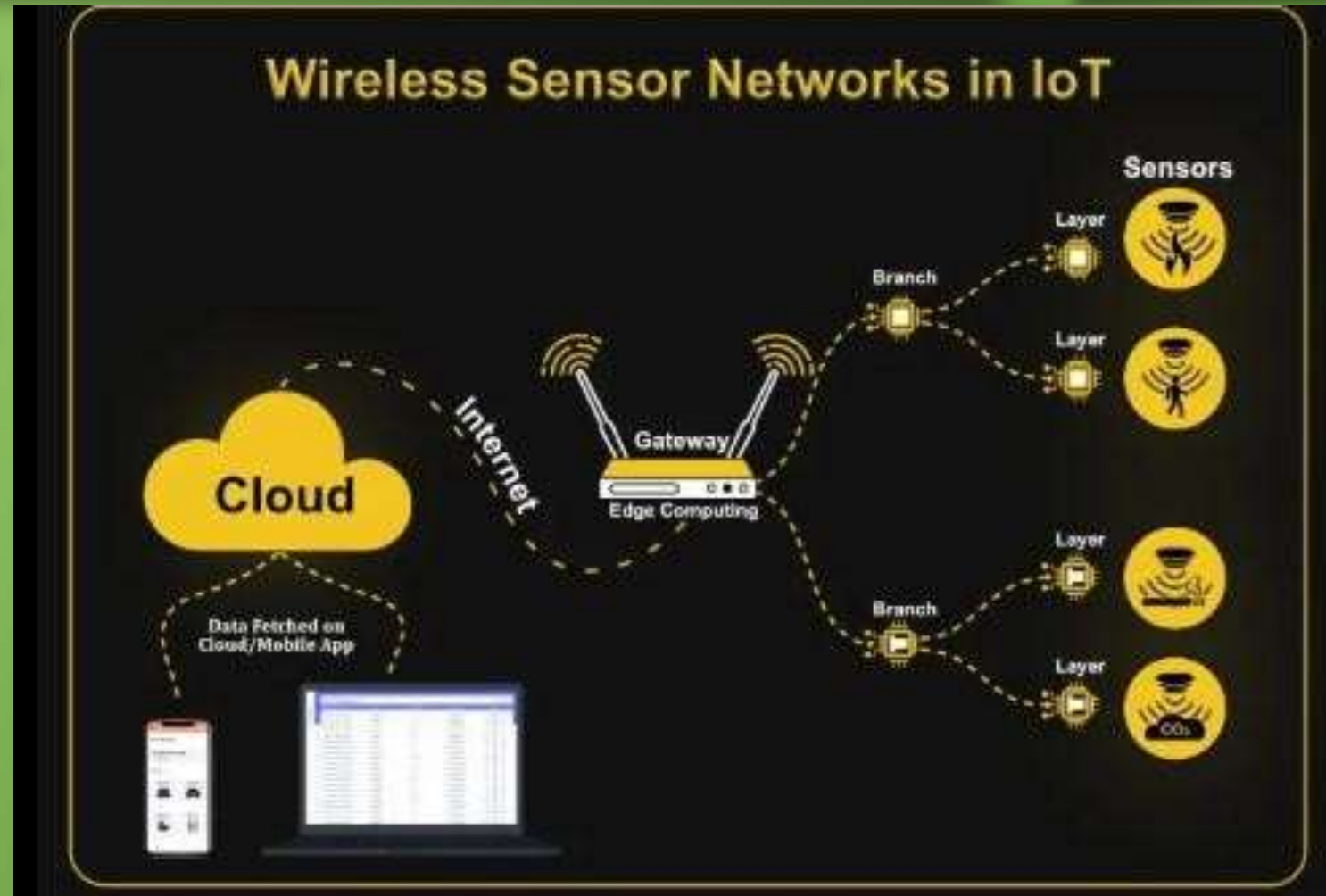
- Toll collection (e.g., FASTag).
- Attendance systems.
- Inventory tracking in warehouses.
- Asset management in hospitals.

2. Sensor Networks (Wireless Sensor Networks - WSN):

- A sensor network is a collection of small electronic devices (sensors) that collect data from the physical world (like temperature, pressure, motion, etc.) and transmit it wirelessly.

Components:

- Sensors
- Microcontrollers
- Wireless communication modules
- Power source

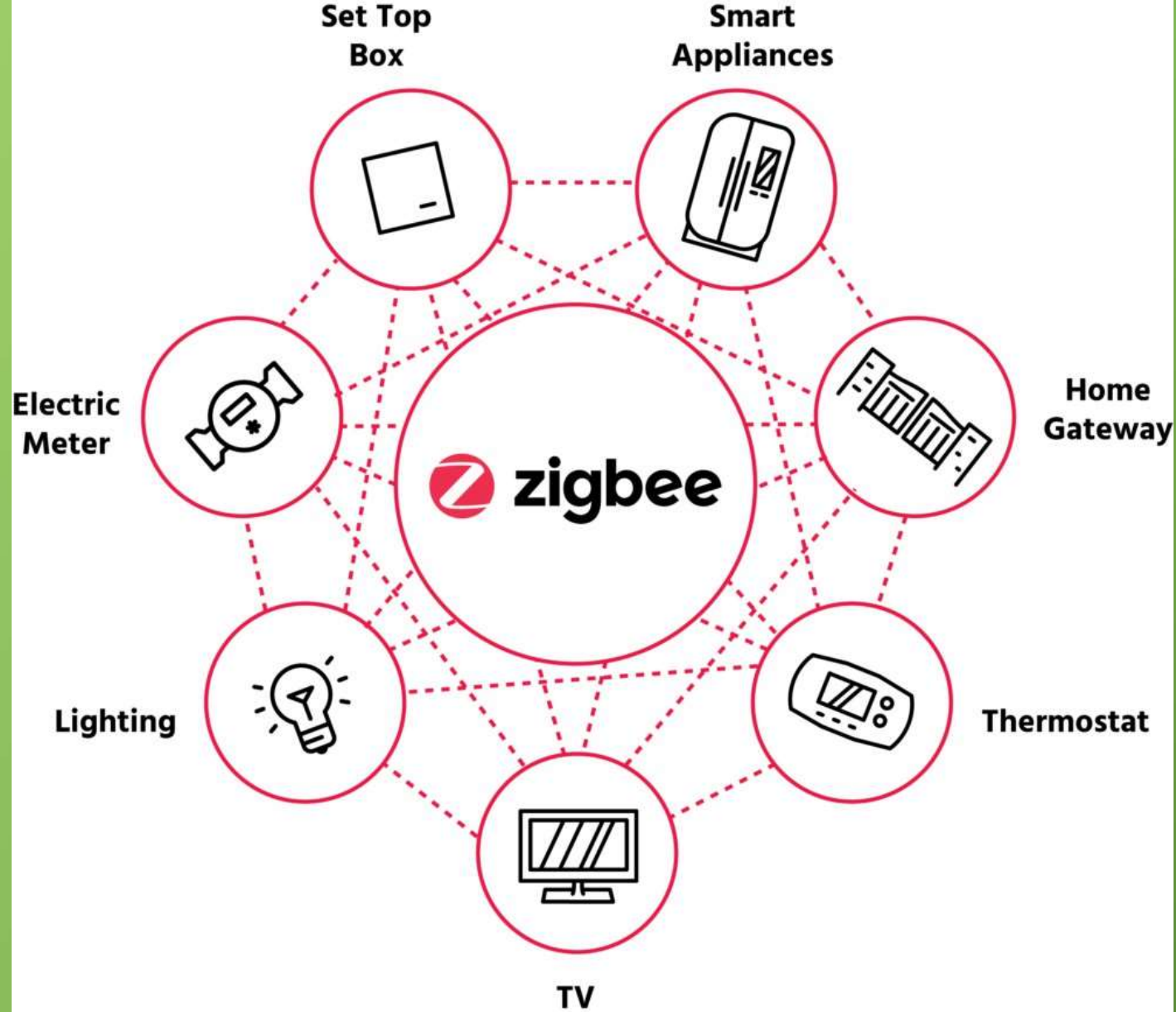


Features and Applications

- **Features:**
 - Real-time data collection.
 - Wireless communication.
 - Low power consumption.
 - Can be deployed in remote areas.
- **Applications:**
 - Smart agriculture (soil/moisture monitoring).
 - Health monitoring (wearable devices).
 - Environmental monitoring.
 - Disaster detection (earthquake sensors).

ZigBee Technology:

- ZigBee is a low-power wireless communication technology used for short-distance data transfer in IoT applications.
- IEEE 802.15.4 standard.



Smart Home

Features:

- Operates in 2.4 GHz frequency.
- Low data rate (250 kbps).
- Mesh network support (devices can relay messages).
- Secure and energy-efficient.
- Range: ~10 to 100 meters.

Applications:

- Smart homes (lighting, fans, ACs).
- Smart energy meters.
- Home security systems.
- Industrial automation.

4. GPS (Global Positioning System):

GPS is a satellite-based navigation system that provides location, time, and velocity information anywhere on Earth.

Components:

- GPS satellites
- GPS receiver (in phone, vehicle, tracker)



Global Positioning System

✓ Features:

- Global coverage.
- Accurate position tracking.
- Real-time navigation.
- Works 24/7 in any weather.

✓ Applications:

- Vehicle tracking.
- Navigation apps (Google Maps).
- Fleet management.
- Personal tracking (wearables).

5.3 Innovative Applications with the Internet of Things

(Ex: Health care: ECG Analysis in Cloud and it's access,
CRM and ERP: Business and Consumer Application)

Questions

- 1) Explain any Home Based application in which Internet of Things is used. (6marks) OR Explain IoT technologies used for home automation and also explain relation of home automation with cloud computing. (6MARKS)
- 2) Describe Home based cloud computing (4M)
- 3) Differentiate between Home Based Cloud Computing and Energy Aware Cloud Computing. (4m)
- 4) List any four innovative applications of cloud with Internet of Things. (4m)
- 5) Enlist technologies in IoT. (Any two)(2m)
- 6) Discuss IoT technologies required for developing smart homes (6M)
- 7) List any four applications of cloud with IOT. (4M)
- 8) Name any two enabling Technologies with the Internet of Things. (2m)
- 9) Explain how cloud can be used to implement IOT Technologies. Explain any one in detail. (6M)

5.3 Innovative Applications with the Internet of Things

The Internet of Things (IoT) enables smart and automated applications in healthcare, business, and consumer sectors. It improves efficiency, accuracy, and real-time decision-making.

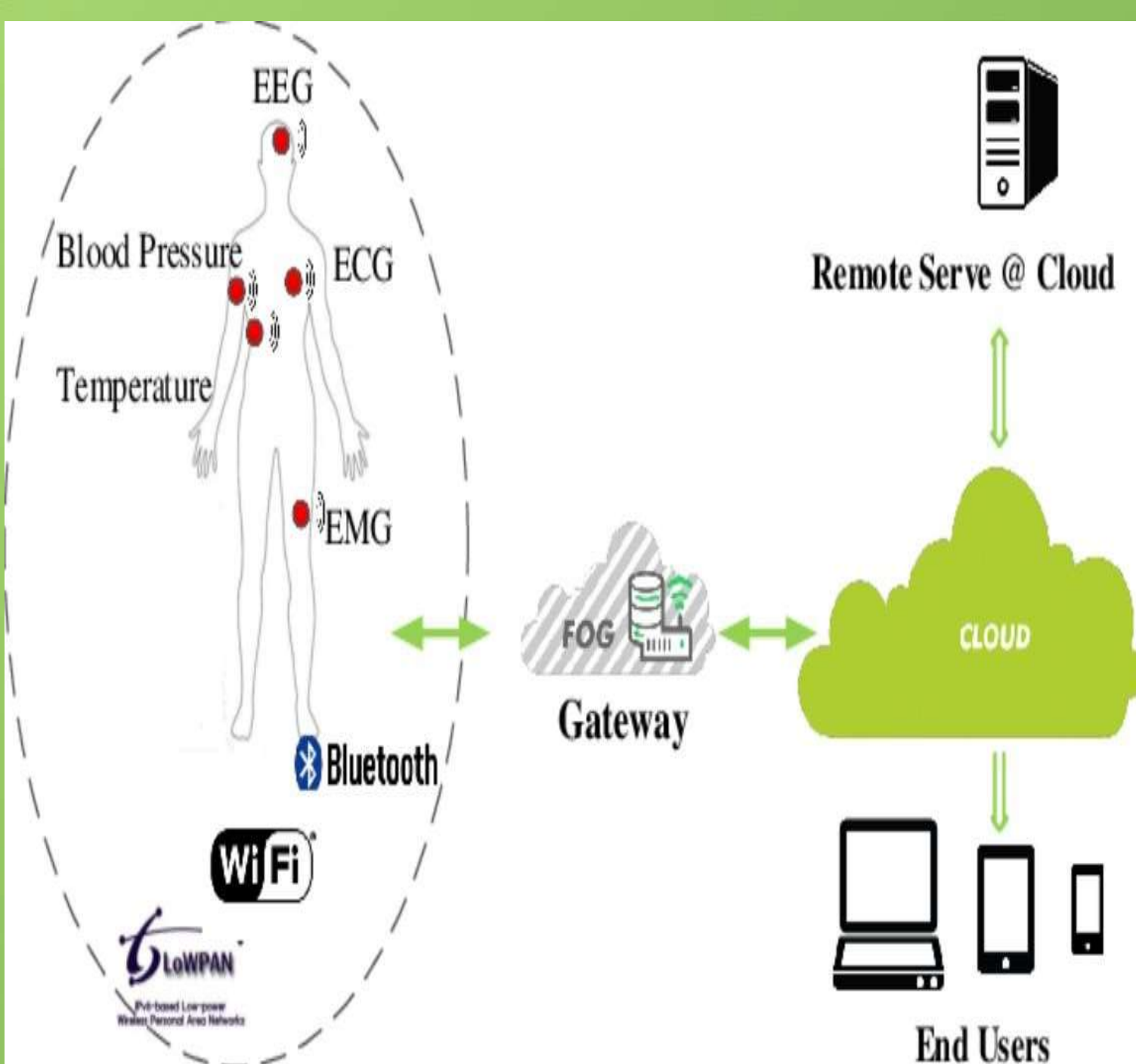
1. Health Care: ECG Analysis in Cloud(JUST FOR EXAMPLE ANY SAME TYPE QUESTION CAN COME)

What is it?

ECG (Electrocardiogram) analysis in the cloud is a **remote patient monitoring system** using IoT.

How it works:

- ECG sensors are attached to a patient's body.
- These sensors collect heartbeat data continuously.
- The data is transmitted to the **cloud using IoT devices** (like smart wearables).
- Doctors can **access this data remotely** using mobile apps or dashboards.



2. CRM (Customer Relationship Management)

- ◆ What is CRM?

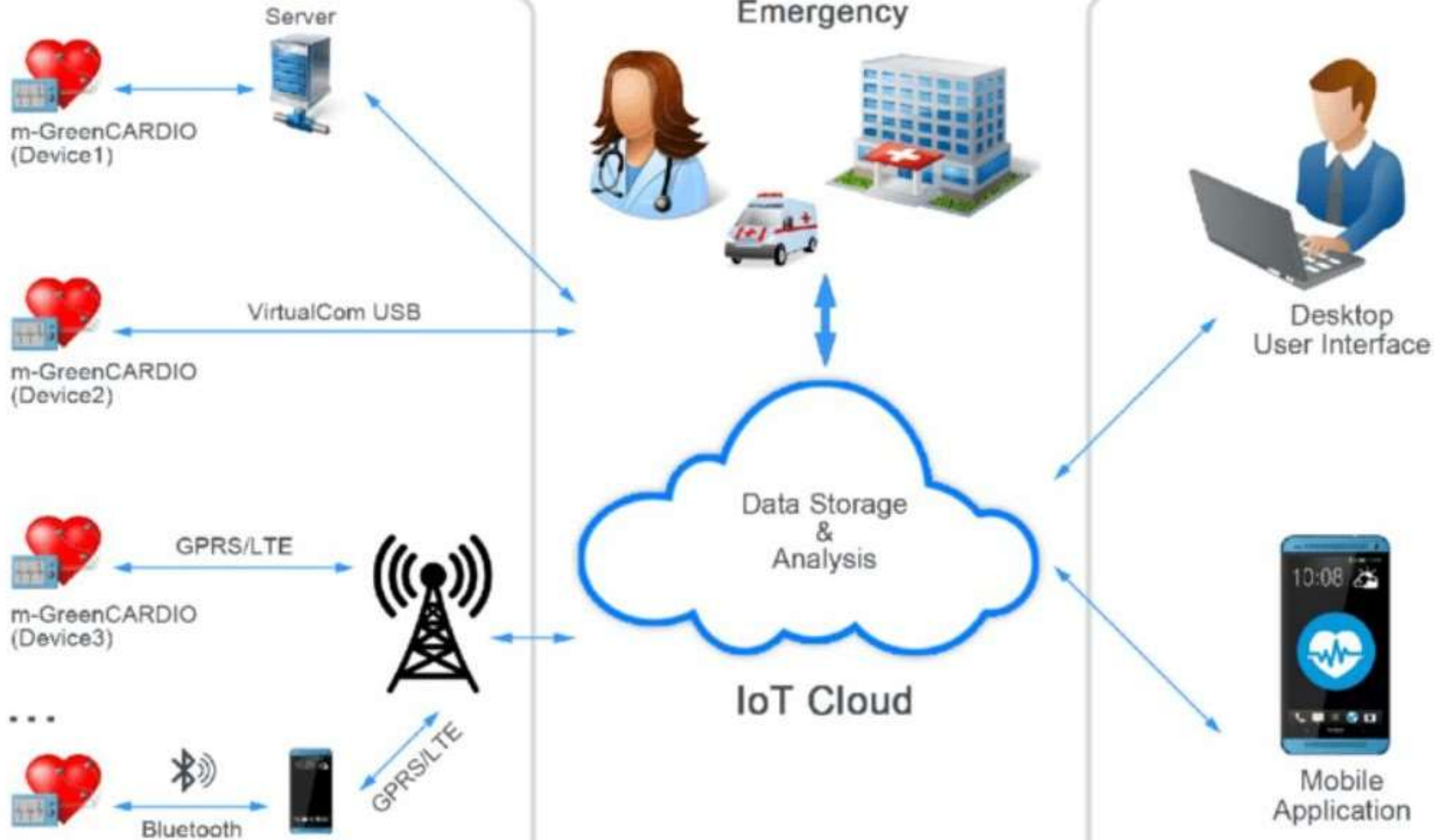
CRM systems help businesses manage customer relationships and interactions.

- ◆ IoT with CRM:

- IoT devices track product usage and customer behavior.
- Data is stored and analyzed by the CRM system.
- Businesses can predict customer needs, send personalized messages, and offer timely service.

- ◆ Example:

Smart ACs send usage data to CRM → Service reminder is sent automatically.



3. ERP (Enterprise Resource Planning)

◆ What is ERP?

ERP is software that integrates all departments of a business into one system.

◆ IoT with ERP:

- IoT sensors track inventory, machine usage, and employee activity.
- Real-time data is sent to the ERP system.
- This helps in automation, resource planning, and cost control.

◆ Example:

Smart factory machines send data to ERP → Predictive maintenance alert is generated → Downtime reduced.

END -----

Questions (2marks)

1. Enlist technologies in IoT (Any two)

- RFID (Radio Frequency Identification)
- Sensor Networks
- ZigBee
- GPS

2. Name any two enabling technologies with the Internet of Things.

- RFID
- ZigBee

Other acceptable answers: Bluetooth, Wi-Fi, NFC, GPS

Questions (4marks)

3. Describe Home Based Cloud Computing

- Home-based cloud computing refers to the use of cloud services and Internet of Things (IoT) devices in a home environment. It allows appliances like smart TVs, lights, ACs, refrigerators, and security cameras to connect to the cloud via the internet.

The cloud enables:

- Remote control of home appliances through smartphone apps
- Storage of usage data and preferences
- Real-time monitoring and automation
- Improved convenience, safety, and energy efficiency

Example:

- A smart air conditioner connected to the cloud can be turned on using a mobile app before the user reaches home. It can also adjust temperature based on past usage data stored in the cloud.

Questions (4marks)

4. List any four innovative applications of cloud with Internet of Things.

- 1.Remote health monitoring (ECG on cloud)
- 2.Smart home control (lights, appliances via app)
- 3.Vehicle tracking with GPS and cloud maps
- 4.Smart farming using cloud-connected sensors

6. List any four applications of cloud with IoT.

- Smart cities (traffic & pollution monitoring)
- Smart healthcare (wearables + cloud storage)
- Smart grids (energy management)
- Smart retail (inventory tracking using cloud + IoT)

Homework!!

1. Differentiate between Home Based Cloud Computing and Energy Aware Cloud Computing (4m)
2. Explain how cloud can be used to implement IoT technologies. Explain any one in detail. (6m)
3. Discuss IoT technologies required for developing smart homes (6m)
4. Explain IoT technologies used for home automation and also explain relation with cloud computing. (6m)
5. Explain any Home Based application in which Internet of Things is used. (6m)

